6.4 Hyperbolas

Name _____ Period ____ Date _____

Pre-Calculus 6.4 · Student sheet

I can…

- write the standard equation of a hyperbola and find its vertices and foci.
- find the asymptotes of a hyperbola.
- classify a conic from its general equation.

Vocabulary · fill in the missing word

Hyperbola — the set of points whose _____ from two fixed points (the foci) have a constant positive difference

Transverse axis — the segment joining the _____

Asymptotes of a hyperbola — the two lines the branches approach, through the corners of the central _____

Part 1 · One sign changes everything

Graph both equations and the lines $y = \pm(2/3)x$ in Desmos. Zoom out.

Equation	Shape	x-intercepts	What happens far away?
$x^2/9 + y^2/4 = 1$			
$x^2/9 - y^2/4 = 1$			

Part 2 · Notes and examples (fill in each note, then work the example)

Standard form (center (h, k)): horizontal $(x - h)^2/a^2 - (y - k)^2/b^2 = 1$; vertical $(y - k)^2/a^2 - (x - h)^2/b^2 = 1$. The _____ term tells the direction.

Foci: c^2 = _____

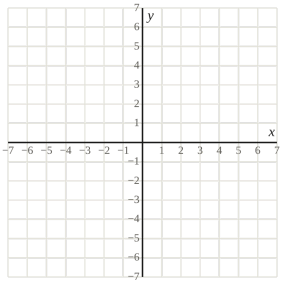
Example 1

Find the vertices and foci of $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

Asymptotes (horizontal transverse axis): $y - k$ = _____$(x - h)$

Example 2

Find the asymptotes of $\frac{x^2}{9} - \frac{y^2}{16} = 1$ and sketch the hyperbola.



Example 3

Write $4x^2 - 3y^2 + 8x + 16 = 0$ in standard form. Find the center, vertices, and foci.

Example 4

Write the equation of the hyperbola with vertices $(0, \pm 3)$ and foci $(0, \pm 5)$.

Classifying $Ax^2 + Cy^2 + Dx + Ey + F = 0$: circle if A = C; parabola if AC = 0; ellipse if AC > 0; hyperbola if AC _____ 0.

Example 5

Classify each conic.

a. $4x^2 - 9x + y - 5 = 0$

b. $4x^2 - y^2 + 8x - 6y + 4 = 0$

c. $2x^2 + 4y^2 - 4x + 12y = 0$

d. $2x^2 + 2y^2 - 8x + 12y + 2 = 0$

Part 3 · Partner practice

Partner 1

Find the vertices, foci, and asymptotes of $\frac{y^2}{25} - \frac{x^2}{144} = 1$.

Partner 2

Classify: a. $3x^2 + 3y^2 - 6x + 9 = 0$ b. $x^2 - 4y^2 + 2x = 0$